INCH-POUND
MIL-PRF-3098/110D
27 July 2005
SUPERSEDING
MIL-PRF-3098/110C
27 August 1997

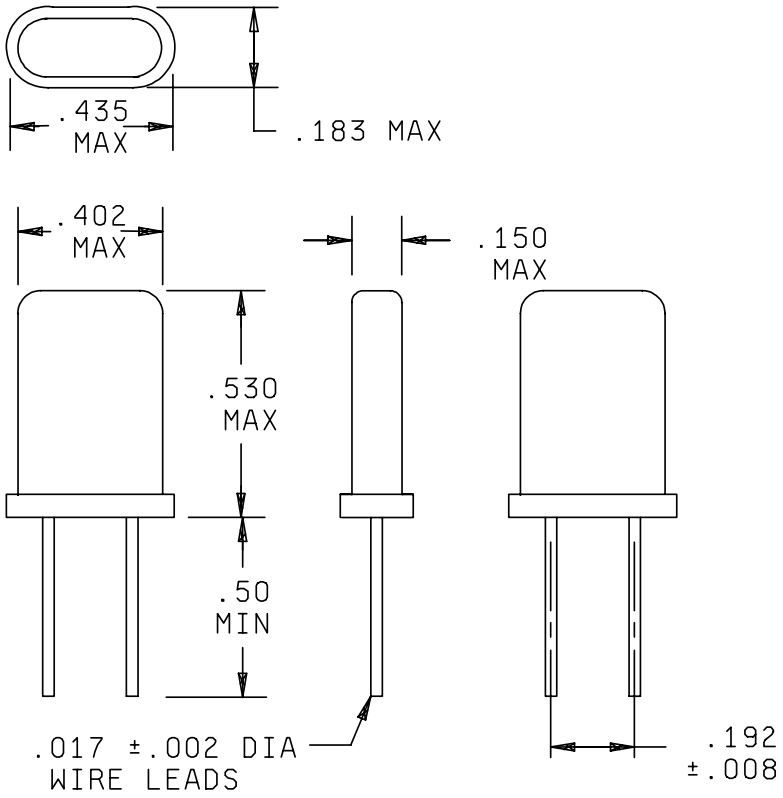
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR129/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 9 MHz to 12 MHz; fundamental; noncontrolled; antiresonance.



Inches	mm
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.500	12.70
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR129/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 9.0 MHz to 12.0 MHz, inclusive.

Capacitance, shunt: 7.0 pF, maximum.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW, maximum.

Antiresonance, load capacitance: 32.0 pF ±0.5 pF.

Operating temperature range (noncontrolled): -55°C to +90°C, inclusive.

Frequency tolerance: ±35 ppm.

Equivalent resistance:

MHz	Ohms, maximum
9.0 to 10.0	35
10.0+ to 12.0	25

Shock (specified pulse): Method 213, MIL-STD-202, test condition K; consisting of 18 impact shocks of 15 gravity units each having a duration of 11 milliseconds; 3 shocks along each direction in each of 3 mutually perpendicular planes.

Frequency change permitted: ±5 ppm.

Equivalent resistance: ±15 percent or 2 ohms, whichever is greater.

Vibration: Method 204, MIL-STD-202; test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent or 2 ohms, whichever is greater.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent or 2 ohms, whichever is greater.

Aging:

Frequency change permitted: ±5 ppm.

* Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

* The margins of this specification sheet are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project 5955-0811)

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